

JAZZ MEDIA CORES

ADVANCED AUDIO PROCESSORS



Digital Radio



Portable Music
Players



Home Audio
Equipment



Surround
Sound Systems



Automotive
Entertainment

High performance cores that enable rapid support of robust audio decoding and encoding.

The Jazz Advanced Audio Processor (AAP) cores are highly optimized solutions specifically tailored to the demanding needs of the quality audio decoding and encoding. Whether you are replacing an existing solution or adding advanced to your products, the Jazz AAP cores can get you to market quickly and efficiently. They support a wide range of audio and speech standards and are optimized for the high quality audio at a low cost. The AAP cores are part of the broad Jazz Media family from Improv.

New digital video devices require support for a variety of different video, audio, image and speech standards. The Jazz AAP solutions support a complete set of audio and speech software modules that can be loaded from external memory; this allows a single core to support a broad range of standards.

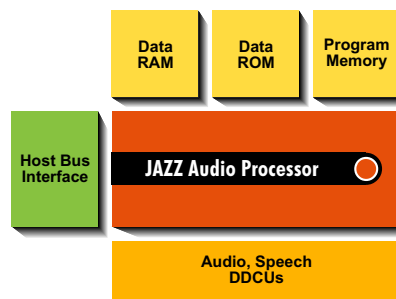
The Jazz AAP cores provide a complete package for supporting a wide variety of broadcast quality audio standards in your product. The cores are pre-configured and pre-optimized and come with a complete suite of audio software modules. The cores have built-in interfaces for standard on-chip buses and memory management to allow direct drop-in into an existing SoC architecture.

OPTIMIZED CORES

The Jazz AAP cores can be used as a black box implementation of complete elementary stream processing for audio processing in a media chip design. Each core can operate as a standalone block or as a slave to a host processor.

Each AAP core has a single, application-specific processor that combines a general-purpose instruction set with application-specific instructions to accelerate audio processing. Each core contains a very-long instruction word (VLIW) processor built using Improv's Jazz Composer tools.

In addition to the processor, the hardware core contains embedded memory, a host bus interface (HBI) that can connect to most standard on-chip buses including AMBA, OCP and Wishbone, and a audio output management Direct Audio Interface (DAI).



KEY FEATURES

- Complete silicon-ready solution
- Optimized for multiple standards at a low cost
- Low clock rates for advanced audio standards
- Available in multiple configurations
- Fully synthesizable and foundry independent
- Supported by a comprehensive software development tool suite
- Optionally available as configurable cores for additional customization

AAP MEDIA SOFTWARE MODULES

The AAP media software modules have been designed to fully exploit the unique advantages of the AAP architecture. The AAP family includes the following modules:

- MPEG L1/L2 Audio Decoder
- MP3 Audio Decoder
- AAC/AAC+ Audio Decoder
- BSAC Audio Decoder
- Windows Media 9 Audio Decoder
- Dolby Digital and ProLogic Audio Decoder
- MIDI Synthesis Engine
- G.711, G.726, G.723.1, G.729a Codecs
- MPEG L1/L2 Audio Encoder
- MP3 Audio Encoder

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AVAILABLE PRODUCTS

The Jazz AAP family consists of the following specific products:

- **AAP Stereo Audio Decoder Core** is optimized for decoding of 2 channel stereo and includes all AAP decoder software modules.
- **AAP Stereo Audio Codec Core** is optimized for decoding and encoding of 2 channel stereo audio and includes all AAP decoder and encoder software modules.
- **AAP Full Surround Audio Decoder Core** is optimized for decoding of 5.1 and 7.1 audio and includes all AAP decoder software modules.
- **AAP Full Surround Audio Codec Core** is optimized for decoding and encoding of 5.1 and 7.1 audio and includes all AAP decoder software modules.

IMPLEMENTATION CHARACTERISTICS

	<u>GATE COUNT</u>	<u>CLOCK</u>
AAP Stereo Decoder	184K gates	25 MHz
AAP Stereo Codec	208K gates	40 MHz
AAP Surround Decoder	184K gates	120 MHz
AAP Surround Codec	208K gates	150 MHz

These numbers represent the maximum requirements for running the AAP Media Software modules in a TSMC 0.13um process. High performance is a critical component of a successful chip for these markets. The Jazz AAP cores have been highly optimized for performance both in audio decoding/encoding and audio processing.

- Fully synchronous design
- Silicon-proven in 0.18um and 0.13um technologies
- Advanced power management features
- Hardware Debug Environment through JTAG port
- Support for prioritized interrupts
- Full scan with BIST for test
- Boot-loadable firmware and smart caching (Jazz Jukebox)
- No custom physical components required

DEVELOPMENT TOOLS

A complete software development environment called the Jazz Standard tool suite supports Jazz cores. This tool suite includes compiler, debugger, assembler, cycle-accurate instruction-set simulator and profiler/analyzer. The environment runs on Windows XP Professional, Windows 2000 and Windows NT environments.

The cores also come with a complete implementation environment called the Jazz PSA Generator which creates complete synthesizable RTL, EDA tool scripts, verification environment and FPGA mapping.

As a separately purchasable item, Improv offers a development platform using Improv's FPGA-Based Jazz Rehearsal™ cards with a full verification environment to enable verification and system integration.

DELIVERABLES

- AAP Processor Core
- AAP Media Software Modules
- Verification Environment
- Jazz Standard Tool Suite
- Jazz PSA Generator
- User's Guide
- Implementation Guide
- Programmer's Guide
- Jazz Composer Tool Suite (for configurable package only)

LICENSING INFORMATION

The Jazz AAP cores are available today and can be licensed from Improv Systems. For additional information regarding the Jazz Media family and other licensable Improv technology, please contact Improv Systems at sales@improvsys.com, or visit our website at <http://www.improvsys.com>. A list of worldwide local representatives can be found on the website.

ABOUT IMPROV SYSTEMS, INC.

Improv Systems, Inc. develops and licenses pre-configured and configurable Jazz DSP, Jazz Media and Jazz Voice. Improv's Jazz processor is a configurable DSP that allows designers to create optimized DSP cores targeted to consumer electronics and telecommunications markets. For more information, please visit www.improvsys.com.

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